

Case 2: $P_2 \rightarrow P_3$. We will give a direct proof.

Let P_2 be True. $\therefore$ Let, $(n-1)$ be an arbitrary odd integer.

By defn 1, $\exists$ an integer K such that $(n-1) = 2K+1$. $\therefore n = 2K+2$

$$n^2 = (2K+2)^2 = 4(K+1)^2 = 2(2K^2 + 4K + 2) = 2\alpha, \text{ where } \alpha = 2K^2 + 4K + 2.$$

$\therefore K$ is an integer, it's easy to see that α is an integer.

By defn 1, n^2 is even. $\therefore P_3$ is True

Case 3: To prove $P_3 \rightarrow P_1$, we will use proof by contraposition.

$\therefore$ We will prove $\neg P_1 \rightarrow \neg P_3$. Let $\neg P_1$ be True. $\therefore$ Let n be an arbitrary integer which is not ~~even~~ ^{odd}. By defn 1, n is ~~even~~ ^{odd}. $\therefore$ By defn 1, there exists an integer K such that $n = 2K+1$. $\therefore n^2 = (2K+1)^2 = 4K^2 + 4K + 1 = 2(2K^2 + 2K) + 1$. $\therefore K$ is an integer, $2K^2 + 2K$ is an integer. By defn 1, n^2 is an odd integer. $\neg P_3$ is: n^2 is not even. By defn 1, n^2 is odd.

$\therefore \neg P_3$ is True

Counterexamples: In Section 1.4, we stated that to show that a statement of the form $\forall x P(x)$ is false, we need only find a counterexample, that is, an example x for which $P(x)$ is false.

Example 15: Show that the statement "Every positive integer is the sum of the squares of two integers." is false.

Solution: Let $x^2 + y^2 = 3$. x and y be integers.

$$x=0 \quad y^2 = 3 - 0^2 = 3 \quad y = \pm\sqrt{3} \text{ which is not an integer}$$

$$x=1 \quad y^2 = 3 - 1^2 = 2 \Rightarrow y = \pm\sqrt{2} \text{ which is not an integer}$$

$$x=2 \quad y^2 = 3 - 2^2 = -1. \text{ There does not exist any integer whose square is negative. For any } x \geq 2, y \text{ as an integer doesn't exist (Easy to prove)}$$